Database Technology - Assignment 6

Note: To solve below queries, use “sales” database

1. Write a query that counts the number of salespeople registering orders for each day. (If a salesperson has more than one order on a given day, he or she should be counted only once.).

```
mysql> SELECT odate, COUNT(DISTINCT snum) AS total_salespeople FROM orders GROUP BY odate;
+------------+--------------------+
| odate      | total_salespeople |
+------------+--------------------+
| 1990-10-03 |                  4 |
| 1990-10-04 |                  3 |
+------------+--------------------+
2 rows in set (0.003 sec)
```

2. Write a query on the Customers table that will find the highest rating in each city. Put the output in this form: For the city (city), the highest rating is: (rating).

```
mysql> SELECT CONCAT('For the city ', city, ' the highest rating is: ', MAX(rating)) AS MAXRATING
FROM customers GROUP BY city;
+-------------------------------------------------+
| MAXRATING                                       |
+-------------------------------------------------+
| For the city London the highest rating is: 100  |
| For the city Rome the highest rating is: 200    |
| For the city San Jose the highest rating is: 300 |
| For the city Berlin the highest rating is: 300  |
+-------------------------------------------------+
4 rows in set (0.001 sec)
```

3 Write an SQL query to calculate the total order amount for each day and display the results in descending order of total orders.

```
mysql> SELECT odate, SUM(amt) AS totalOrderAmount FROM orders GROUP BY odate ORDER BY totalOrderA mount DESC;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server
version for the right syntax to use near 'mount DESC' at line 1
mysql> SELECT odate, SUM(amt) AS totalOrderAmount FROM orders GROUP BY odate ORDER BY totalOrderAmount DESC;;
+------------+------------------+
| odate      | totalOrderAmount |
+------------+------------------+
| 1990-10-04 |         16713.81 |
| 1990-10-03 |          8944.59 |
+------------+------------------+
2 rows in set (0.001 sec)
```

4. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the average amount of the order in the table.
(Note Use the average amount value directly →2565.84)

```
mysql> SELECT snum, SUM(amt) AS totalAmount FROM orders GROUP BY snum HAVING SUM(amt) > 2565.84;;
+------+-------------+
| snum | totalAmount |
+------+-------------+
| 1001 |    15382.07 |
| 1002 |     5546.15 |
+------+-------------+
2 rows in set (0.001 sec)
```

5. Write a query that selects the highest rating in each city.

```
mysql> SELECT city, MAX(rating) AS highSal FROM customers GROUP BY city;
+----------+---------+
| city     | highSal |
+----------+---------+
| London   |     100 |
| Rome     |     200 |
| San Jose |     300 |
| Berlin   |     300 |
+----------+---------+
4 rows in set (0.001 sec)
```

6. Retrieve the maximum order amount for each salesperson from the orders table, including only those orders whose value exceeds Rs. 3000.

```
mysql> SELECT snum, MAX(amt) AS maxOrderAmt FROM orders WHERE amt > 3000 GROUP BY snum;
+------+-------------+
| snum | maxOrderAmt |
+------+-------------+
| 1002 |     5160.45 |
| 1001 |     9891.88 |
+------+-------------+
2 rows in set (0.001 sec)
```

7. Write an SQL query to find the smallest order placed by every customer.

```
mysql> SELECT cnum, MIN(amt) AS smallOrder FROM orders GROUP BY cnum ORDER BY cnum;
+------+------------+
| cnum | smallOrder |
+------+------------+
| 2001 |     767.19 |
| 2002 |    1713.23 |
| 2003 |    5160.45 |
| 2004 |      75.75 |
| 2006 |    4723.00 |
| 2007 |    1900.10 |
| 2008 |      18.69 |
+------+------------+
7 rows in set (0.001 sec)
```